When Verifiers Backfire: A Cost–Correctness Theory of Generation–Verification–Repair Loops for LLMs

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Abstract

Test-time verification is becoming a standard component of LLM agent systems: a generator produces an artifact, a verifier accepts or rejects it, and rejections trigger repair. We give a cost–correctness theory of this generation–verification–repair (GVR) loop—including a correction of a tempting over-reading of it—and stress both in a two-benchmark study with full silent-oracle instrumentation: one generator (qwen-plus), six LLM judge configurations spanning three model families, unit-test oracles, budget N=5, 164×9 HumanEval and 247×9 MBPP runs, every attempt ground-truth-scored. On the theory side, misses hurt final precision $R = a/(1-b)$ times as much per unit as false rejections when retries redraw at the generator's rate; degraded retries ($q < q^*$) flip that ranking; and—the correction—per-event dominance of false rejections does not by itself imply backfire: the loop backfires iff its delivered-correct probability A_N falls below the deliver-first-attempt baseline, equivalent in the steady state to $R_q < (1-a)/b$. Empirically the backfire is real, severe, and sharply patterned by judge configuration. Same-family judges lose correct work—net -15 correct solutions for the isolated self-judge on HumanEval ($p = .0007$), -10 for the cross-capability judge on both benchmarks—while neither cross-family configuration (deepseek-v4-pro, glm-5.2) shows a reliable negative effect, and glm-5.2 trends positive on MBPP (+8, nominal $p = .039$). A same-family reasoning-judge control dissociates the axes: the collapse signature persists within the family, but the significant losses are confined to the non-reasoning judges. The mechanism is not broken repair: retries succeed at $q = .47$–$.95$, but same-family judges' acceptance of correct work collapses across rounds (round 1: .75–.87; round ≥ 4: .00–.36) although the judges cannot see the round—the empirical signature of instance-persistent verifier–generator correlation, one of two failure regimes our theory separates; the other, priced by the flip condition, governs the cross-capability arm, and the corrected condition, evaluated at the experiment's budget, signs all twelve judge-benchmark cells. Oracle verification never loses a correct solution (+20/+17 rescues on MBPP, $p < 10^{-4}$; leak exactly zero wherever b=1), and fully paired feedback and verifier-context ablations yield no statistically reliable effect. The engineering lesson is sharp: whether a loop purifies or grinds is governed by the verifier's configuration relative to the generator—family-associated in our arms—and its acceptance persistence on resubmitted work—measure both before trusting any verifier.

1 Introduction

Production LLM agent systems increasingly wrap generation in a verification loop: an executor produces an artifact, a verifier judges it, rejections trigger repair, and the loop terminates when the verifier accepts or the budget runs out. This generation–verification–repair (GVR) pattern underlies executor/ verifier pipelines, LLM-as-judge filters, and test-driven agent loops. The engineering question it raises is sharp and often ignored: *when does adding a verifier actually improve the final answer, and when does it make things worse?*

Two literatures border this question without answering it: the self-correction literature documents that intrinsic self-correction frequently fails while sound external verification helps (Stechly et al., 2025; Kamoi et al., 2024; Huang et al., 2024), and the test-time-scaling theory of Dorner et al. (2026) covers pure resampling—

rejection carries no feedback, retries are independent redraws, and no operating-point guidance is given. Neither analyzes the repair loop with feedback.

This paper contributes a compact theory of the GVR loop and a two-benchmark, nine-arm study (164×9 HumanEval; 247×9 MBPP) with one generator, six LLM judge configurations from three model families, and silent-oracle instrumentation of every attempt. Our contributions:

1. **A corrected backfire theory (§3.2–3.5).** The miss-asymmetry exchange rate $R = a/(1-b)$; its degradation modulation R_q and per-event flip threshold q^* (Theorem 5); and the terminal backfire condition (Corollary 3): the loop backfires iff $A_N(p, a, b, q) < p$, equivalent in the steady state to $R_q < (1-a)/b$. Per-event dominance of false rejections does not by itself imply backfire—a verifier whose catches rescue enough wrong work stays above the baseline even with $q < q^*$.

2. **Configuration-dependent backfire, family-associated in our arms (§4).** Under identical tasks, prompts, generator budget, and instrumentation, same-family judges lose correct work (isolated self-judge -15 on HumanEval, $p = .0007$; cross-capability judge -10 on both benchmarks), while neither cross-family configuration shows a significant negative effect and glm-5.2 trends positive on MBPP ($+8$, nominal $p = .039/.012$). On HumanEval the cross-family judges are near-oracles (leak .012 at $\hat{a} + \hat{b} \approx 1.7$). The cross-family judges are also stronger, reasoning models—a co-varying confound a dedicated control arm dissects: the same-family judge served with thinking enabled and the cross-family verdict budget still collapses and clusters, yet nets $-3/-4$ (n.s.), so the significant backfires require the same-family correlation *and* the non-reasoning operating point.

3. **The mechanism: acceptance collapse, not broken repair (§4).** Retries succeed at $q = .47$–$.95$, yet same-family judges' acceptance of correct attempts collapses with round ($\hat{a}_1 \approx .75$–$.87$, $\hat{a}_{k\geq 4} \approx .00$–$.36$) although the isolated judges cannot see the round, and their rejections of correct work cluster by task—evidence consistent with the instance-persistent correlation that Theorem 4 prices. The terminal backfire condition, evaluated at the experiment's budget, signs all twelve judge-benchmark cells; the pooled two-stage model flags the self-judge arms only marginally and underpredicts their losses by an order of magnitude—the gap is the collapse itself.

4. **Nulls and instrumentation (§4).** A fully paired feedback ablation (shared failing code, McNemar) detects no feedback-content effect ($p = .50/.55$ on the two benchmarks), leaving $\delta > 0$ (Proposition 1) undetermined; a verifier-context probe again finds no collapse; single-shot verification purifies the delivered stream in all 12 judge-benchmark cells; and completion-cap instrumentation matters—a 1,024-token pilot truncated 36–68% of round-2 completions by arm on HumanEval into unparsable code, masquerading as repair degradation.

2 Related Work

Self-correction and its critiques. Self-Refine (Madaan et al., 2023) and Reflexion (Shinn et al., 2023) popularized iterative self-improvement; Stechly et al. (2025) documented verifier false-rejection pathologies, with Huang et al. (2024) and the critical survey of Kamoi et al. (2024) reaching compatible conclusions. Our theory gives these observations a probabilistic form: collapse corresponds to failure-mode correlation between generator and verifier (§3.3), and our experiment delimits where that correlation appears.

Test-time scaling theory. Dorner et al. (2026) prove that rejection sampling and Best-of-N accuracy is determined by verifier ROC geometry, and that RS dominates BoN at fixed compute. Their setting is pure resampling: no feedback, no repair, no generator–verifier correlation analysis, and no operating-point selection guidance. Our paper is a deliberate three-axis extension: repair feedback (§3.4), operating-point calibration (§3.2), and degraded/correlated retries (§3.5, 3.6). Related theory includes inference-time alignment coverage analysis (Huang et al., 2025). In the training-time regime, Plesner et al. (2026) study the robustness of RLVR training to noisy verifiers and report that verifier precision matters more than recall; our Theorem 3 supplies the inference-time counterpart of that asymmetry in closed form.

Verification techniques. Self-consistency (Wang et al., 2023) uses agreement as the verification signal; process supervision (Lightman et al., 2024) trains step-level verifiers; LLM-as-judge evaluation is surveyed by Zheng et al. (2023). Control-theoretic formalizations of self-correction dynamics appear in Wu et al. (2026); they manage loop stability, whereas we manage the verifier's error budget. All of these techniques select a point on the verifier's quality trade-off; our results say *which* point to select and when the loop tolerates a harsh verifier.

3 Theory

Table 6 (Appendix A) summarizes the paper's notation.

3.1 Setup: the GVR loop

One *attempt*: a generator produces an artifact; a verifier either *accepts* or *rejects* it (with feedback); a rejection triggers a new attempt, up to budget N.

Definition 1 (Per-attempt rates)**.** *Let correctness of an attempt be binary, with generator success $p = \Pr(C)$. The verifier has acceptance rate $a = \Pr(\textit{accept} \mid C)$ and catch rate $b = \Pr(\textit{reject} \mid \neg C)$. Its two error types are the* miss $1 - b$ *(accepting wrong work) and the* false rejection $1 - a$ *(rejecting correct work).*

Definition 2 (Attempt outcomes)**.** *Per attempt: $s = pa$ (accept-correct), $\ell = (1-p)(1-b)$ (*leak*: accept-wrong), and $r = 1 - s - \ell$ (reject).*

Terminal states: *accepted* (split into correct and leak) and *unresolved* (budget exhausted). We assume (A1) the verdict is independent of the generation path given correctness and (A2) attempts are i.i.d. with constant (p, a, b); both are relaxed in §3.4–3.6.

Lemma 1 (Single-pass composition)**.** *For $N = 1$, $\Pr(\textit{deliver wrong}) = (1-p)(1-b)$: a wrong deliverable requires the conjunction of two independent failures. The false-rejection rate does not enter this quantity.*

Proof. Direct computation of $\Pr(\neg C \wedge \text{accept})$. □

Theorem 1 (Loop closed forms)**.** *Under (A1)–(A2), with budget $N \geq 1$:*

$$\Pr(\textit{leak}) = \ell \, \frac{1 - r^N}{1 - r}, \quad \Pr(\textit{accepted}) = (s + \ell) \, \frac{1 - r^N}{1 - r}, \tag{1}$$

$$\Pr(\textit{unresolved}) = r^N, \qquad \mathbb{E}[\textit{attempts}] = \frac{1 - r^N}{1 - r}. \tag{2}$$

Proof. The loop is geometric: reaching attempt k requires $k - 1$ rejections (probability r^{k-1}); summing and telescoping gives the claims. The instance-level structure matches the ROC-based characterization of Dorner et al. (2026) for the pure resampling case; we state it in outcome-decomposed form for use below. □

Theorem 2 (Budget–precision invariance)**.** *For every $N \geq 1$,*

$$\pi := \Pr(\textit{wrong} \mid \textit{accepted}) = \frac{(1-p)(1-b)}{pa + (1-p)(1-b)},$$

independent of N: added budget converts unresolved mass into accepted mass at fixed precision.

Proof. Ratio of the leak and accepted probabilities in Theorem 1; the geometric factors cancel. □

3.2 The miss-asymmetry theorem

Both verifier errors hurt π—misses first-order through ℓ, false rejections second-order: each forced retry buys a fresh chance to leak. Their marginal damages have a closed-form ratio.

Theorem 3 (Miss-asymmetry)**.**

$$\frac{\partial\pi/\partial(1-b)}{\partial\pi/\partial(1-a)} = \frac{a}{1-b}.$$

Proof. $\partial\pi/\partial(1-b) = (1-p)s/(s+\ell)^2$ and $\partial\pi/\partial(1-a) = p\ell/(s+\ell)^2$ by the chain rule on Definition 2; the ratio simplifies to $a/(1-b)$. □

At $(a, b) = (0.95, 0.95)$, $R \approx 19$: one unit of catch is worth ∼19 units of acceptance for precision, while $1-a$ enters expected attempts first-order and $1-b$ only through the negligible ℓ term.

Corollary 1 (Calibration guideline)**.** *Under (A2), any infinitesimal verifier modification—or one small enough that the operating point moves little—trading acceptance for catch at a local exchange rate better than* 1:R *improves final precision to first order; its price is paid only in attempts. Calibrate verifiers aggressively toward catch rate; budget false rejections rather than fear them. The qualifier is essential: when retries are degraded below* q^* *(Theorem 5) the effective exchange rate drops below 1 and the advice inverts—protect acceptance.*

Corollary 2 (Majority voting)**.** *Replace a verifier by a majority vote of* k *conditionally independent verifiers (strict majority rejects; ties accept). For* $a, b > 1/2$ *the vote raises* both *effective rates:*

$$b^{(k)} = \sum_{j>k/2} \binom{k}{j} b^j (1-b)^{k-j} > b, \qquad a^{(k)} = 1 - \sum_{j>k/2} \binom{k}{j} (1-a)^j a^{k-j} > a,$$

because majority rule makes false rejections rarer*, not commoner—a lone objecting judge no longer suffices to reject. Precision improves through both channels, and generator attempts do not increase (r falls); the price is* k *verifier calls per round (majority-of-three at the pooled self-judge point: generator attempts* $1.38 \to 1.25$*, verifier calls* $1.38 \to 3.75$*, steady state under (A2),* $< 1\%$ *change at* $N{=}5$*). Unanimity-style rules (reject if any judge objects) trade the other way:* $a^{(k)} = a^k$ *falls sharply while* $b^{(k)} = 1-(1-b)^k$ *rises.*

3.3 Correlated failures

(A1) fails in a specific direction: a verifier that is the same model, primed with the generator's context, tends to repeat the generator's mistake. Model this with a latent failure-mode mixture.

Definition 3 (Failure-mode mixture)**.** *Conditioned on latent mode* $M = m$ *with prior* π_m*, generation succeeds with* p_m*; the verifier accepts correct output with* a_m *and catches wrong output with* b_m*, independently within mode.*

Definition 4 (Effective rates)**.** *With* $\epsilon = \sum_m \pi_m(1-p_m)$*, error mass* $\mu_m = \pi_m(1-p_m)/\epsilon$ *and success mass* $\sigma_m = \pi_m p_m/(1-\epsilon)$*:* $b_{\text{eff}} = \sum_m \mu_m b_m$ *is the error-mass-weighted catch rate and* $a_{\text{eff}} = \sum_m \sigma_m a_m$ *the success-mass-weighted acceptance rate.*

Theorem 4 (Decorrelation condition)**.** *Attempt-1 outcomes obey the closed forms of §3.1 with pooled rates* $(p, a, b) \to (1-\epsilon,\, a_{\text{eff}},\, b_{\text{eff}})$*; in particular the accepted-stream error*

$$\pi_1 = \frac{\epsilon\,(1-b_{\text{eff}})}{\epsilon\,(1-b_{\text{eff}}) + (1-\epsilon)\,a_{\text{eff}}}$$

improves on no verification (delivered error ϵ*) iff*

$$a_{\text{eff}} + b_{\text{eff}} > 1,$$

i.e., iff the mass-weighted operating point lies above the chance diagonal of ROC space: verification helps exactly when $\Pr(\textit{accept} \mid C) > \Pr(\textit{accept} \mid \neg C)$*. Catch alone does not suffice: if* $b_{\text{eff}} \to 0$ *while* $a_{\text{eff}} < 1$*, verification is* strictly harmful *(*$\pi_1 = \epsilon/[\epsilon + (1-\epsilon)a_{\text{eff}}] > \epsilon$*), degenerating to neutral theater only in the rubber-stamp limit* $a_{\text{eff}} \to 1$*. Correlated failure has an exact form: with* $\nu = \sum_m \pi_m(1-b_m)$ *the prior-weighted miss rate,*

$$\Pr(\textit{wrong} \wedge \textit{miss}) = \epsilon(1-b_{\text{eff}}) = \epsilon\nu + \text{Cov}_m(1-p_m,\, 1-b_m),$$

the covariance over modes under prior π_m*: decorrelation is precisely the claim that this covariance be small.*

For $N > 1$ the mixture bites: with instance-persistent modes, surviving to round k conditions on the mode mix. §4 validates the attempt-1 form directly and then observes the mixture in the raw: judges that cannot see the round nonetheless accept correct work at collapsing per-round rates as rounds accumulate—the signature of instance-persistent low-a_m modes conditioning the surviving pool.

3.4 Repair feedback breaks i.i.d. — favorably

Proposition 1 (Repair factor)**.** *Let rejected attempts be followed by a repaired attempt succeeding with $p' = p + \delta(1-p)$, $\delta \in [0,1]$ ($\delta = 0$: rejection sampling; $\delta = 1$: perfect repair). Then all formulas of §3.2 with constant p are conservative (actual leak is upper-bounded), and the invariance of Theorem 2 breaks in the favorable direction: precision improves with budget.*

Proof sketch. Sample a uniform u once per attempt slot and couple a repaired draw (correct iff $u \le p'$) with a fresh one (correct iff $u \le p$); since $p' \ge p$, the repaired draw is correct whenever the fresh one is. Every round $k \ge 2$ therefore has masses $s' = p'a \ge s$ and $\ell' = (1-p')(1-b) \le \ell$, so its accepted stream is at least as pure, $\pi' = \ell'/(s'+\ell') \le \pi$, and every constant-$p$ formula is an upper bound on realized leak. The terminal conditional error at budget N is a weighted mean of π (round 1) and π' (rounds ≥ 2) with the π'-weight strictly increasing in N; it therefore decreases toward π'—the favorable break of Theorem 2's invariance. □

The bound isolates the repair channel: the verifier's (a, b) are held fixed, and repair is assumed to act on generator success alone—it does not speak to verifier–generator correlation (§3.3).

3.5 Repair degradation flips the asymmetry

Real repairs do not redraw at the generator's first-attempt rate: the retry pool is conditioned on rejection, enriched for hard instances, and its success rate q can fall far below p—mildly for some judges in our study, severely for oracle-side repair on the harder benchmark.

Definition 5 (Retry rate)**.** $q := \Pr(\textit{attempt } k{+}1 \textit{ correct} \mid \textit{attempt } k \textit{ rejected})$, $k \ge 1$.

A falsely rejected *correct* solution enters a sub-loop with per-round success q; conditional on resolution it ends wrong with

$$\pi_q = \frac{(1-q)(1-b)}{qa + (1-q)(1-b)}.$$

Theorem 5 (Modulated exchange rate)**.** *At the population level, the marginal damage ratio becomes*

$$R_q \;=\; \frac{a}{1-b} \cdot \frac{q(1-p)}{p(1-q)}.$$

The asymmetry flips *(false rejections dominate) iff*

$$q < q^* \;=\; \frac{(1-b)p}{(1-p)a + (1-b)p}.$$

Proof. A unit ε of extra miss rate moves $\varepsilon(1-p)$ mass of wrong first attempts from "caught" (sub-loop wrong with probability π_q) to "delivered wrong" (probability 1): damage $\varepsilon(1-p)(1-\pi_q)$. A unit of extra false-rejection rate moves εp mass of correct first attempts from "delivered correct" to the sub-loop: damage $\varepsilon p \pi_q$. Substituting $\pi_q = \ell_q/(s_q + \ell_q)$ with $s_q = qa$, $\ell_q = (1-q)(1-b)$ and taking the ratio yields R_q; setting $R_q = 1$ and solving for q gives q^*. For $q = p$ the expression recovers Theorem 3. □

Corollary 3 (Terminal backfire condition)**.** *In the two-stage model (round 1 at p, retries at q, constant (a, b)), the delivered-correct probability at budget N is*

$$A_N \;=\; pa \;+\; r_1\, qa\, \frac{1 - r_q^{N-1}}{1 - r_q}, \qquad r_1 = p(1-a) + (1-p)b, \qquad r_q = q(1-a) + (1-q)b,$$

and the loop backfires relative to the deliver-first-attempt baseline iff $A_N < p$. In the infinite-budget limit this is equivalent to

$$R_q \;<\; \frac{1-a}{b}\,:$$

terminal backfire requires the modulated exchange rate to fall below $(1-a)/b$—for verifiers above the chance line ($a+b>1$, hence $(1-a)/b<1$), a strictly deeper drop than the per-event flip at $R_q=1$. Per-event dominance of false rejections does not by itself imply backfire: a verifier whose catches convert enough wrong first attempts into rescued correct work can remain above the baseline even with $q<q^*$. The conversion rate of correct solutions into wrong ones remains $(1-a)\,p\,\pi_q$.

Proof. Round 1 delivers correct mass pa; reaching sub-round $k \geq 2$ requires the round-1 rejection mass r_1 followed by $k-2$ sub-loop rejections (r_q each) and delivers qa; summing the geometric series gives A_N. For the limit, $A_\infty < p$ iff

$$r_1\,qa < p(1-a)\big[qa + (1-q)(1-b)\big] \iff (1-p)\,b\,q\,a < p\,(1-a)\,(1-q)\,(1-b),$$

and substituting R_q from Theorem 5 turns the right-hand condition into $R_q < (1-a)/b$. Since $A_N < A_\infty$ for every finite N, the steady-state condition is sufficient for backfire at any budget (finite N only enlarges the backfire region). □

3.6 What pooled rates cannot see

The pooled $(\hat{a}, \hat{b})$ reported by evaluation studies summarize attempts across instances and rounds. Three phenomena hide beneath the summary: (i) retry degradation (§3.5) multiplies per-retry leak chance from $(1-p)(1-b)$ to $(1-q)(1-b)$; (ii) misses cluster on particular wrong-solution instances that recur across rounds, so instance-level modes with low b_m carry disproportionate mass; and (iii) acceptance can collapse on particular *correct* instances—modes with low a_m that survive every round—which the pooled $\hat{a}$ averages away. §4 measures (iii) directly, uses (i) to bound (ii), and shows that the magnitude of pooled-rate failure is itself instrumentation-sensitive: pooled theory understates terminal leak only mildly in our clean runs, and a completion-cap artifact can inflate the apparent failure substantially.

4 Experiments

Setup. Two benchmarks: HumanEval (Chen et al., 2021), all 164 problems, and MBPP (Austin et al., 2021), 247 tasks from the hand-verified subset of the official test split (task IDs 11–510). Generator: qwen-plus (temperature 1.0), prompted to plan then emit a Python block; budget N=5. Nine arms: none (deliver first attempt); oracle_full / oracle_verdict (unit tests in a timeout-limited subprocess; feedback = failing-test traceback / verdict only); and six LLM judge configurations—judge_hetero (qwen-max), judge_self_iso and judge_self_ctx (qwen-plus, the generator's own family; the latter sees the generator's reasoning), two cross-family judges, judge_deepseek_iso (deepseek-v4-pro) and judge_glm_iso (glm-5.2), and a same-family reasoning-judge control, judge_self_iso_thinking (qwen-plus with thinking enabled, 4,096-token verdict budget), added to dissect the family/capability confound. Every attempt is additionally scored by the oracle *silently*, so per-arm operating points, per-round dynamics, leak composition, and terminal outcomes are all measured directly.

Instrumentation note. Completion budgets matter twice over. *Generator side*: a pilot run at a 1,024-token completion cap truncated 36–68% of round-2 completions by arm on HumanEval (30–45% on MBPP; first attempts: 4–6% and 0–2%) into unparsable code, which masqueraded as severe repair degradation; the reported runs use an 8,192-token cap, under which unparsable rates fall to 0.8% (HumanEval) and 0.2% (MBPP), with a per-attempt parse-health flag recorded. *Judge side*: the non-reasoning Qwen judges emit short verdicts, and their 512-token verdict budget was never binding (zero verdict-format failures across all same-family cells); the thinking control's 4,096-token budget likewise never bound (two MBPP tasks were lost to repeated API read timeouts and are excluded from its statistics, n=245 of 247); the cross-family reasoning judges, given 4,096, occasionally exhaust that budget mid-thought, and the harness's conservative fallback records these as rejections (38/28 events for glm-5.2 and 4/7 for deepseek-v4-pro on

Table 1: HumanEval main results (n=164 per arm; 95% Wilson CIs). π_{pred} from Theorem 2 with measured $\hat{p}_1, \hat{a}, \hat{b}$; π_{obs} = error conditional on delivery.

Arm	$\hat{p}_1$	$\hat{a}$	$\hat{b}$	Acc.	Leak	Unres.	$\mathbb{E}[k]$	π_{pred}	π_{obs}
none	.95	–	–	.95	.049	.000	1.00	–	–
oracle_full	.95	1.00	1.00	.99	.000	.006	1.07	.000	.000
oracle_verdict	.96	1.00	1.00	.99	.000	.012	1.10	.000	.000
judge_hetero	.95	.68	.71	.89	.043	.067	1.46	.022	.046
judge_self_iso	.93	.58	.78	.84	.043	.116	1.64	.026	.048
judge_self_ctx	.95	.71	.81	.91	.024	.067	1.41	.015	.026
judge_deepseek_iso	.95	.88	.89	.97	.012	.018	1.21	.006	.012
judge_glm_iso	.94	.80	.87	.95	.012	.037	1.27	.011	.013
judge_self_iso_thinking	.95	.72	.83	.93	.018	.049	1.40	.012	.019

Table 2: MBPP main results (n=247 per arm, thinking control 245; conventions as Table 1).

Arm	$\hat{p}_1$	$\hat{a}$	$\hat{b}$	Acc.	Leak	Unres.	$\mathbb{E}[k]$	π_{pred}	π_{obs}
none	.74	–	–	.74	.259	.000	1.00	–	–
oracle_full	.74	1.00	1.00	.82	.000	.178	1.86	.000	.000
oracle_verdict	.74	1.00	1.00	.81	.000	.190	1.89	.000	.000
judge_hetero	.74	.77	.41	.70	.263	.036	1.36	.213	.273
judge_self_iso	.73	.72	.56	.69	.239	.073	1.51	.185	.258
judge_self_ctx	.73	.75	.45	.71	.235	.053	1.38	.215	.248
judge_deepseek_iso	.74	.82	.33	.76	.235	.008	1.28	.225	.237
judge_glm_iso	.73	.88	.38	.77	.223	.012	1.23	.207	.225
judge_self_iso_thinking	.73	.66	.55	.71	.233	.057	1.59	.203	.247

HumanEval/MBPP, predominantly on correct attempts)—an asymmetry that biases *against* the cross-family arms, which nonetheless dominate; where we isolate the judge's own verdict behavior (Table 4) we exclude these events. The API reports served identifiers qwen-plus, qwen-max, deepseek-v4-pro, glm-5.2 (rolling aliases; October 2026). Throughout, an attempt is *correct* iff it passes the benchmark's test suite (the silent oracle); suite adequacy is a limitation (§5).

4.1 Oracle verification never loses correct work

Unit-test verification behaves exactly as the theory demands. Leak is .000 in all four oracle arm-benchmark cells (Theorem 1 at b=1), no correct solution is ever re-rolled away, and the terminal-vs-own-first-attempt contrast is uniformly positive: +7/+5 rescues on HumanEval ($p = .016/ .063$; the wrong-first pools are only 7–8 tasks) and +20/+17 on MBPP (both $p < 10^{-4}$). MBPP also displays the budget side of the trade: with repair genuinely hard there ($\hat{q} \approx .17$), the oracle arms convert extra budget into unresolved mass (18–19%) rather than wrong deliveries. At $a = 1$ the backfire condition is vacuous; at $b = 1$ the single-shot stream is exactly pure.

4.2 Same-family judges lose correct work; cross-family judges do not

Cross-arm comparisons are confounded by first-draw sampling, so the causal comparison pairs each arm's terminal outcome against *its own* first attempt on the same task (Table 3).

Table 3 is the paper's cleanest contrast, and two regularities organize it. First, *every same-family judge cell is negative, and the four significant negative cells are all same-family*: the isolated self-judge loses net −15 on HumanEval ($p = .0007$) and −10 on MBPP ($p = .021$); the cross-capability judge—still Qwen-family—loses −10 on both benchmarks ($p = .006$ each). Second, *no cross-family cell is ever significantly negative*: deepseek-v4-pro trends positive on both (+3/+5, n.s.), and glm-5.2 is *positive* on MBPP (+8, nominal $p = .039$; $p_{\text{del}} = .012$). On HumanEval the cross-family judges are near-oracles; on MBPP they are lenient but harmless

Table 3: Within-arm pairing: terminal outcome vs. the same arm's own first attempt (McNemar exact). "Lost"/"rescued": first attempt correct/terminal not, and first wrong/terminal correct; p: unresolved as failure (primary); p_{del}: unresolved excluded; effect in accuracy points.

Arm	Lost	Rescued	Net	p	p_{del}	Effect
HumanEval (n=164 per arm)						
oracle_full	0	7	+7	.016	.016	+4.3pp
oracle_verdict	0	5	+5	.063	.063	+3.0pp
judge_self_iso	17	2	−15	.0007	1.00	−9.1pp
judge_hetero	11	1	−10	.006	.625	−6.1pp
judge_self_ctx	9	3	−6	.146	.250	−3.7pp
judge_deepseek_iso	1	4	+3	.375	.125	+1.8pp
judge_glm_iso	3	5	+2	.727	.063	+1.2pp
judge_self_iso_thinking	8	5	−3	.581	.219	−1.8pp
MBPP (n=247 per arm; thinking control n=245)						
oracle_full	0	20	+20	$< 10^{-4}$	$< 10^{-4}$	+8.1pp
oracle_verdict	0	17	+17	$< 10^{-4}$	$< 10^{-4}$	+6.9pp
judge_self_iso	13	3	−10	.021	1.00	−4.0pp
judge_hetero	11	1	−10	.006	.219	−4.0pp
judge_self_ctx	10	6	−4	.454	.289	−1.6pp
judge_deepseek_iso	3	8	+5	.227	.109	+2.0pp
judge_glm_iso	2	10	+8	.039	.012	+3.2pp
judge_self_iso_thinking	10	6	−4	.455	.508	−1.6pp

(Tables 1, 2). The harmful configuration is exclusively the middle one: a judge harsh enough to reject correct work ($\hat{a} = .58$–$.77$) and, as §4.3 shows, persistent about it.

The same-family losses are not delivered errors but *churned* correct work: of the self-judge's 17 HumanEval losses, 16 end unresolved and only one is delivered wrong; MBPP shows the same shape (13 losses: 4 delivered-wrong, 9 unresolved). The conversion channel that the theory prices, $(1 - a)p\pi_q$, has collapsed (HumanEval conversions across the six judges: 3/1/0/0/0/1), because retry quality is no longer binding: repairs succeed at $\hat{q} = .47$–$.95$. What fails is not the repair but the verifier's willingness to accept it. The churn shape also explains the p_{del} column: excluding unresolved tasks removes exactly the tasks the verifier destroyed, so a delivered-only convention cannot witness churn-type losses by construction.

The judge configurations co-vary—the cross-family judges are also stronger, *reasoning* models—and the judge_self_iso_thinking control dissects the confound: it serves the same qwen-plus family with thinking enabled and the cross-family 4,096-token verdict budget. On the family side it behaves like its siblings—its acceptance still collapses and clusters (§4.3)—but on the outcome side it does not: net −3 on HumanEval ($p = .58$) and −4 on MBPP ($p = .45$), against −15/−10 for its non-reasoning sibling. The significant backfires therefore require the same-family correlation *and* the non-reasoning operating point, and the difference lives in resubmission behavior: on MBPP the control accepts 62% of correct round-2 attempts against the plain self-judge's 42%, so the churn never becomes a significant loss. The remaining budget asymmetry is inert and biased against the winners: the non-reasoning Qwen judges' 512-cap was never binding, while the cross-family reasoning judges occasionally exhausted their own budgets mid-thought, and the harness recorded those events as rejections concentrated on correct attempts (excluding them, glm's acceptance of correct work is .96 on HumanEval and .94 on MBPP, against .80/.88 as measured inside the loop); verdict costs differ widely (medians: 240–860 tokens non-reasoning Qwen, 860–1,480 cross-family, ≈ 3,500 control), but the only asymmetry that exists points against the cross-family arms. We report nominal p-values throughout: under Benjamini–Hochberg control across the twelve judge cells, the HumanEval self-judge and both cross-capability backfires remain significant, while the MBPP self-judge ($p = .021$) and glm's positive effect ($p = .039$) do not—we read both as strongly suggestive. The pattern itself is complete sign separation: the eight same-family cells are exactly the eight most-negative effects of the twelve (exchangeability $p = 1/\binom{12}{8} \approx .002$).

4.3 The mechanism: per-round acceptance collapse

The same-family judges' pooled operating points look merely mediocre ($\hat{a}$ = .58–.77). The per-round instrumentation shows what the pool hides (Table 4): acceptance of *correct* attempts collapses as rounds accumulate. The isolated self-judge accepts 82% of correct first attempts on HumanEval but 24% of correct second attempts and 0–22% thereafter (MBPP: same shape; Table 4). The cross-capability and context judges collapse similarly; late-round cells are individually noisy (n = 5–29) but the trajectory is monotone and replicates. The thinking control collapses in family fashion (.87 → .45 → .20 on HumanEval, .75 → .62 → .35 on MBPP), though less steeply than its non-reasoning siblings. No cross-family judge collapses: on parsed verdicts (*Instrumentation note*), deepseek-v4-pro on MBPP accepts correct work at .84 (round 1) and .88 (round 2), and glm-5.2's apparent HumanEval decline vanishes—its round-2 acceptance among parsed verdicts is 12/12.

The isolated judges cannot see the round—their input is the problem and the code, nothing else—so the collapse cannot be a response to round information. It is also not a harness artifact in the arms where it appears: the same-family judges never exhausted their verdict budgets (zero format failures), so every same-family rejection in Table 4 is a parsed verdict. Surviving-pool composition differs across arms, but the generator and retry prompts are shared, and the cross-family judges accept the same retry stream at a flat rate—neither code style nor instrument explains a same-family-specific pattern. A round-blind judge that accepts fresh correct work at .8+ and survivors near zero is responding to *which tasks* survive: each rejection round enriches the surviving pool for instance-persistent low-a_m modes, exactly the correlated-failure mixture of Theorem 4. This is how same-family backfire persists at high repair quality: repairs succeed ($\hat{q}$ = .47–.95), but correct repairs on hostile-mode tasks are re-rejected until the budget dies.

The rejections also cluster by task, as the mixture predicts. On HumanEval the isolated self-judge rejected *every* correct attempt on 19 of the 32 tasks where it saw more than one; its per-task rejection counts are overdispersed relative to a pooled binomial (χ^2/df = 1.31); the thinking control clusters like its siblings (χ^2/df = 1.03, with 8 of 24 multi-correct tasks all-rejected). The cross-family arms show no such clustering on either benchmark (χ^2/df = .60–.72; deepseek-v4-pro on MBPP rejected all correct attempts on 0 of the 26 tasks where it saw more than one). Selection makes this diagnostic imperfect—a task must be rejected to keep producing rounds—so we read it as evidence *consistent with* instance persistence rather than an identification of it; the contrast—same-family clustered, cross-family diffuse, on both benchmarks—is what the theory anticipates.

The terminal-backfire condition (Corollary 3) prices the pooled parameters, and Table 5 shows both of its forms against the data. The exact finite-budget form, $A_5 < \hat{p}_1$, signs all twelve cells: it flags the four significant backfires and clears the eight cells that are not, the thinking control included (under a pure sign labeling it additionally misses the four negative-but-nonsignificant context and control cells). The cross-capability arms are flagged clearly ($-2.5/-1.4$pp predicted against $-6.1/-4.0$pp observed), because their retry quality is genuinely degraded on the hard pool ($\hat{q}$ = .71/.51 against $\hat{p}_1$ = .95/.74, below q^* under either rejection-pool estimate)—the degradation regime of Theorem 5. The self-judge arms are flagged only marginally (-0.1pp each, against $-9.1/-4.0$pp observed), and the residual is the point: pooling averages away the acceptance collapse of Table 4, which is exactly the difference between the model's -0.1pp and the measured losses. The steady-state reduction $\hat{R}_q < (1-\hat{a})/\hat{b}$ is the interpretable limit and is sufficient-only—it agrees everywhere except the two self-judge cells, which the finite-budget truncation term catches. The two theorems are complementary: the flip condition (Theorem 5) prices degradation-driven backfire, the mixture (Theorem 4) prices correlation-driven backfire, and the experiment measures both.

4.4 Single-shot purification and the budget

Truncating every run at budget N and recomputing conditional-on-delivery error traces the purity curve (Table 7 in the appendix). At N=1, all twelve judge-benchmark cells are purer than their baselines—exactly the chance-line prediction of Theorem 4, since every pooled operating point satisfies $\hat{a} + \hat{b} > 1$. What the added budget buys depends on the configuration. For the cross-family judges the stream barely moves; for the non-reasoning same-family judges purity decays with budget as churned mass accumulates (the control decays only mildly, .007 → .019 HumanEval and .230 → .247 MBPP, below baseline throughout),

Table 4: Per-round acceptance of correct attempts among *parsed* verdicts, $\hat{a}_k = \Pr(\text{accept} \mid \text{correct}, \text{round } k)$ (cells with $n \geq 5$; round-1 cells have $n \geq 146$). Cross-family cells exclude verdict-format fallbacks (Setup), so all cells measure the judge's own verdicts. The isolated judges see only the problem and the code—they cannot see the round.

Arm	k=1	k=2	k=3	k=4	k=5
HumanEval					
judge_self_iso	.82	.24	.22	.00	.05
judge_hetero	.81	.52	.47	.00	.00
judge_self_ctx	.86	.55	.17	.08	.11
judge_glm_iso	.97	1.00	–	–	–
judge_deepseek_iso	.93	.71	–	–	–
judge_self_iso_thinking	.87	.45	.20	.36	.12
MBPP					
judge_self_iso	.86	.42	.31	.12	.00
judge_hetero	.86	.55	.22	.22	.20
judge_self_ctx	.86	.56	.45	.00	.10
judge_glm_iso	.95	.89	.80	–	–
judge_deepseek_iso	.84	.88	.60	–	–
judge_self_iso_thinking	.75	.62	.35	.25	.25

Table 5: The terminal-backfire condition (Corollary 3) against observed within-arm net effects. $\hat{q} = \hat{p}_2$ (mixed rejection pool). Primary prediction: the finite-budget form $A_5 < \hat{p}_1$ (in-sample consistency check, not an out-of-sample forecast); the steady-state reduction $\hat{R}_q < (1-\hat{a})/\hat{b}$ is shown for interpretation and is sufficient-only. Discrimination: 12/12 under the significance labeling (under a pure sign labeling the four negative, nonsignificant context and control cells are additionally missed).

Arm	$\hat{q}$	$A_5 - \hat{p}_1$	Pred.	$\hat{R}_q$	$(1-\hat{a})/\hat{b}$	Net (significance)
HumanEval						
judge_hetero	.71	−2.5pp	yes	.30	.45	−10 (p=.006)
judge_self_iso	.83	−0.1pp	yes	.92	.53	−15 (p=.0007)
judge_self_ctx	.81	+1.9pp	no	.95	.36	−6 (n.s.)
judge_deepseek_iso	.83	+3.9pp	no	2.03	.14	+3 (n.s.)
judge_glm_iso	.95	+5.0pp	no	7.83	.23	+2 (n.s.)
judge_self_iso_thinking	.79	+1.5pp	no	.81	.33	−3 (n.s.)
MBPP						
judge_hetero	.51	−1.4pp	yes	.48	.57	−10 (p=.006)
judge_self_iso	.47	−0.1pp	yes	.54	.50	−10 (p=.021)
judge_self_ctx	.64	+3.1pp	no	.91	.56	−4 (n.s.)
judge_deepseek_iso	.72	+3.5pp	no	1.15	.54	+5 (n.s.)
judge_glm_iso	.65	+4.6pp	no	.94	.33	+8 (p=.039)
judge_self_iso_thinking	.66	+4.1pp	no	1.09	.61	−4 (n.s.)

and the cross-capability arm on MBPP—lowest catch ($\hat{b} = .41$) and harshest churn—is the only cell whose delivered stream crosses the no-verification baseline (at N=3: .267 vs. .259). On HumanEval no arm crosses at any budget, though the self-judge approaches it by N=5. Budget is thus not neutral (Theorem 2 assumes generator-rate retries).

4.5 Pooled rates: a mild, family-associated gap

Pooled operating points understate observed terminal leak only mildly in the clean runs. Per-arm ratios $\pi_{\text{obs}}/\pi_{\text{pred}}$ are 1.19–2.12 on HumanEval (geometric mean 1.72, bootstrap 95% CI $[0.90, 3.33]$; leak counts are only 2–7 per arm, and 22% of resamples contain a zero-leak arm) and 1.05–1.39 on MBPP (geometric mean 1.19, $[0.95, 1.43]$; both same-family arms individually exclude 1). A degradation-only model at the

measured per-round success trajectory reproduces the same-family gaps in full on MBPP (106–109% of the pooled-to-observed gap), so no miss-clustering channel is required there; the cross-family arms' tiny gaps are not separately explained (72–75%) but sit within noise of zero.

Two lessons. First, the *magnitude* of pooled-rate failure is instrumentation-sensitive: the same three HumanEval judge arms that measure 2.1/1.8/1.7 here measured 3.4/3.3/3.6 under the pilot's 1,024-token cap. Second, pooled $\hat{a}$ hides the acceptance collapse of §4.3 in the opposite direction: the pool says the self-judge accepts 58% of correct work; the round-2 truth is 24% and falling. Pooled rates are blind to the two per-round structures that drive the loop.

4.6 Feedback content: a paired null ablation

The within-run comparison of traceback vs. verdict-only feedback is confounded by independent first draws, so we re-ran the ablation *paired*: for every task whose oracle-arm first attempt failed, the same failing code was re-submitted to two repair sub-loops differing only in feedback content (verdict-only vs. full traceback; same generator, temperature, and budget; McNemar exact on repair success). The result is a clean null: 5/8 vs. 7/8 repaired on HumanEval ($p = .50$), 18/64 vs. 21/64 on MBPP ($p = .55$), pooled discordants 4 vs. 9 ($p = .27$)—the direction consistently favors traceback but is far from significance, and the within-run (unpaired) comparison is likewise null. We conclude that $\delta > 0$ (Proposition 1) remains open here: feedback content is not detectably the lever at this sample size.

4.7 No context-induced collapse

The collapse motivation of §3.3 predicts that a judge shown the generator's own reasoning should be *worse*. On both benchmarks the context-sharing judge sits between the isolated self-judge and the cross-family arms: a slightly better operating point, per-round acceptance still collapsing (Table 4), and a non-significant within-arm deficit (-6, $p = .15$; -4, $p = .45$). We find no evidence of context-induced collapse and cannot exclude moderate harm; what is new here is that the judge-configuration contrast dwarfs the context contrast: correlation travels with the judge configuration, not the prompt—it must be measured (Theorem 4), not assumed.

5 Discussion

Design guidelines. **G1: oracle-first.** Wherever a deterministic check exists (tests, compilers, schemas), attach it to the verifier: leak is exactly zero in every oracle cell we measured, at bounded cost. **G2: mind the configuration, not just the operating point.** The cheapest lever we measured is judge selection: no cross-family configuration showed a significant backfire, and on HumanEval the cross-family judges approached oracle behavior, while same-family judges with similar operating points destroyed correct work. The thinking control shows both axes act—the collapse signature travels with the family, significant damage requires the weak non-reasoning configuration—so prefer a strong verifier from another family; as a heuristic that held in every cell we measured. **G3: probe acceptance persistence.** Take correct solutions the judge has rejected and resubmit them: if acceptance keeps falling by round (Table 4's pattern), the correlated-failure mode is active—change the verifier, not the prompt. **G4: check the flip condition where repair is genuinely hard.** When retries succeed well below the generator's rate (our cross-capability arm; oracle repair on MBPP at $\hat{q} \approx .17$), Corollary 1's catch-first advice inverts—protect acceptance (Corollary 3). **G5: never trust pooled operating points.** They average away both per-round structures—retry degradation and acceptance collapse (§4.5); report per-round health. **G6: budget is a verifier property.** Single-shot purification held in all twelve judge cells (§4.4); every same-family degradation and the only baseline crossing arrived with budget, a correlated verifier converting extra budget into extra rounds of re-rejecting the same correct work.

Limitations. One generator (qwen-plus) at one temperature; judges are one model per family from three families, via rolling aliases (API identifiers recorded); cross-family arms are isolated-context only, so the family and context levers are not fully crossed. The family/capability confound is dissected by the thinking

control but not dissolved to single ingredients: with one model per family and one reasoning toggle, whether it is reasoning per se, verdict budget, or another capability ingredient that mutes the damage remains open—though the control's collapse shows the correlation channel itself is family-associated. Two code benchmarks with unit-test oracles: correctness is measured relative to the benchmark test suites, not exhaustively, so "leak" means suite-passing wrong work. Same-family losses concentrate in unresolved tasks, so delivered-error composition rests on small conversion counts (≤ 5 per arm on HumanEval); late-round acceptance cells rest on $n = 5$–29 attempts and are individually noisy, though the monotone collapse replicates. The acceptance-persistence statistics are survivor-selected—consistent with, not identifying, instance-persistent modes. The paired feedback ablation (§4.6; 72 pairs) is underpowered for small δ; p-values are nominal (FDR control: §4); and q is estimated from mixed rejection pools (the correctly-rejected-pool estimate exceeds q^* for every arm but the cross-capability one).

6 Conclusion

Verification is not free reliability, and its failure modes have structure: a verifier can hurt through degraded retries (priced by the flip threshold q^*) or through correlation with the generator (exposed by per-round acceptance), and pooled operating points hide both. In our study the second mode dominated: non-reasoning same-family judges churned correct work to death while cross-family judges sat near the oracle—and the thinking control shows the signature alone does not suffice. The practical summary: *before trusting a verifier, measure what it does to correct work it has already rejected once.*

Reproducibility. Both experiments use qwen-plus as generator (temperature 1.0, completion cap 8,192 tokens) with judges qwen-max and qwen-plus (512-token verdict budget), deepseek-v4-pro and glm-5.2 (4,096; reasoning models), and a qwen-plus reasoning control (enable_thinking, 4,096), served via a DashScope-compatible API (October 2026); budget N=5; bootstrap seed 20261016. Complete per-attempt logs (1,476 HumanEval runs, 2,221 MBPP runs, 10.7M tokens; every attempt silently oracle-scored with parse health), 144 paired-ablation variants, prompts, code, and analysis scripts are in the supplementary material.

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A Notation

Table 6: Notation.

Symbol	Meaning
p, p_k	generator success rate per attempt; round-k conditional success
a, b	verifier operating point: Pr(accept $\mid C$), Pr(reject $\mid \neg C$)
$1-a$, $1-b$	false rejection; miss (the two verifier error types)
s, ℓ, r	per-attempt accept-correct, leak, and rejection masses (Definition 2)
N, $\mathbb{E}[k]$	attempt budget; expected attempts
π, π_q	conditional-on-delivery error of the loop; of the retry sub-loop
q, q^*	retry success rate (Definition 5); flip threshold (Theorem 5)
R, R_q	miss-asymmetry exchange rate (Theorem 3); its degraded-retry modulation
δ	repair factor (Proposition 1)
ϵ, a_{eff}, b_{eff}	mixture error mass and effective rates (Definitions 3–4)

B Budget sweep

Table 7: Conditional-on-delivery error π by budget N (runs truncated at N rounds). No-verification baselines (deliver the first attempt unverified): .049 (HumanEval), .259 (MBPP).

Arm	N=1	N=2	N=3	N=4	N=5
HumanEval (baseline .049)					
judge_hetero	.023	.048	.046	.046	.046
judge_self_iso	.031	.036	.035	.042	.048
judge_self_ctx	.029	.027	.026	.026	.026
judge_deepseek_iso	.007	.006	.006	.013	.012
judge_glm_iso	.014	.013	.013	.013	.013
judge_self_iso_thinking	.007	.020	.020	.019	.019
MBPP (baseline .259)					
judge_hetero	.245	.250	.267	.268	.273
judge_self_iso	.214	.237	.242	.258	.258
judge_self_ctx	.244	.244	.246	.249	.248
judge_deepseek_iso	.240	.233	.237	.239	.237
judge_glm_iso	.225	.226	.225	.222	.225
judge_self_iso_thinking	.230	.227	.242	.246	.247